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Wind-AE: A Python package for atmospheric escape with metals and X-rays

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Software

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Summary

Throughout their lives, close-in exoplanets are strongly irradiated by their host stars. In the first few hundred million years of life, though, atomic species in a planet's upper atmosphere are photoionized by the young hot stars' strong X-ray and extreme UV radiation. This ionizing radiation from the star heats the planets' upper atmospheres to up to 10,000 K and causes the gas to expand, accelerate from sub- to supersonic speeds (a.k.a., a Parker wind), and outflow from the planet – essentially photoevaporating part or all of the planet's atmosphere in a process called atmospheric escape. Understanding the atmospheric mass loss histories of these planets is essential to understanding the evolution of exoplanets close to their host stars. However, mass loss rates are not directly observable. They can only be inferred from models. To that end, we have developed `Wind-AE`: a fast, 1D, steady-state Parker wind forward relaxation model based on (Murray-Clay et al., 2009) that can model irradiation by multifrequency X-ray/extreme-UV (XUV) stellar spectra and the interactions of these high energy photons with "metals" (atomic species).

Statement of Need

`Wind-AE` is a Python-wrapped and significantly updated version of the widely-cited closed-source (Murray-Clay et al., 2009) C code. The updates to the model physics primarily include the ability to model multifrequency XUV photons interacting with metals in the planets' upper atmospheres. Modeling impact of the presence of metals and X-rays is important because the two have the ability to dramatically change the mass loss rates and wind radial structure for certain planets. `Wind-AE` is unique for its combination of X-ray/metal physics and speed. Speed is of particular interest to users in the broader exoplanet communities for whom `Wind-AE` presents a variety of uses including as a simple plug-and-play model that quickly returns mass loss rates allowing and as a model that can be easily integrated with more sophisticated models. There are no non-isothermal models (important for calculating accurate mass loss rates), nor models with self-consistent ionization calculations (important for predicting observability of escaping atmospheres with telescopes) with `Wind-AE`'s speed, a speed it achieves though the use of the numerical analytical relaxation method.

State of the Field

For detailed scientific comparisons of other 1D open source atmospheric escape models, see Appendix A of our paper. Broadly speaking, `Wind-AE` runs faster than many models that perform more detailed physical calculations and makes more physically realistic assumptions than many models of comparable or faster speed.

- ATES is a comparably rapid model that includes X-ray radiation from the star, but differs from `Wind-AE` in that it approximates the stellar spectrum as a power law, only models atomic hydrogen and helium in the atmosphere, and calculates the ionization balance post facto (Caldiroli et al., 2021).
- p-winds is an isothermal Parker wind model which is approximately 10 times faster than `Wind-AE`, but serves a slightly different purpose as users set the mass loss rate to obtain the wind's structure. `p-winds` models extreme-UV radiation (EUV), but does not model X-rays. By solving for the temperature as a function of radius, `Wind-AE` is able to obtain more realistic outflow profiles, but at the cost of speed. ((Dos Santos et al., 2022)).
- Sunbather wraps `p-winds` and Cloudy (a spectral synthesis code (Ferland et al., 2017)) to create transit models that include metals (Linssen et al., 2024).
- TPCI leverages Cloudy and the hydrodynamic code PLUTO ((Mignone et al., 2012)) for more thorough X-ray and EUV-irradiated simulations, but slightly more expensive calculations ((Salz et al., 2015)).
- pyTPCI is a Python wrapper for TPCI (Zhang et al., 2025).
- AIOLOS models diffusion (a physical process whose impact on atmospheric escape is still being investigated) and the lower atmosphere in more detail and in a time-dependent manner, so it takes of order 100 times longer (Schulik & Booth, 2023).
- CETIMB includes more detailed lower atmosphere physics and, therefore, takes of order 500 times longer Taylor et al. (2025).
- CHAIN integrates CLOUDY and (D. Kubyshkina et al., 2018). Like TPCI, via Cloudy, CHAIN has slightly more nuanced heating and cooling calculations and does model X-rays (though the published grid is EUV only (D. I. Kubyshkina & Fossati, 2021)), so is also more expensive.

Software Design

The three main tenants that underly our software design are (1) speed, (2) user-friendliness and minimizing the scientific expertise barrier to entry, and (3) automating and minimizing hands-on time.

(1) `Wind-AE`'s unique advantage is its speed for its level of sophistication. This speed is achieved by using the numerical analysis method called relaxation and the numerical calculations are done in the low-level language, C, to maintain that speed advantage. The original source code used in (Murray-Clay et al., 2009) was developed in 2007 and, as such, uses C, as opposed to C++, C#, or Cython and C is still widely used in astronomy. The C source code computes the "solutions" to the atmospheric escape radial density, temperature, ionization fraction and column density for each atomic species in the wind. For numerical efficiency the computation of these radial values is performed by the source code in uncommon units and the solution files are information heavy, making direct user importing or reading challenging. Thus, we provide a Python wrapper that converts these radial values into an easily accessible Pandas dataframe and also into commonly used astronomy units.

(2) From the above radial values, a variety of valuable diagnostics can be computed postfacto and added to the solution dataframe. These diagnostics include: pressure, heating and cooling rates, ionization rates, collisionality, optical depth, and many many more as a function of radius. Evaluating these diagnostics allows for analysis of a variety of wind properties and has helped users make novel discoveries about, e.g., wind energetics.

The postfacto computation of these diagnostics is handled by a Python wrapper. Python-based user interfaces are far more accessible than C to the majority of astronomers, from the researchers to the undergraduate students who have used this code. Thus, a Python wrapper is critical for the user-friendliness that underlies our software design. To increase user-friendliness and lower the amount of atmospheric expertise users require to use `Wind-AE`, we also include a wide variety of automatic flags that are raised to warn users when, e.g., the `Wind-AE` solution is outside of the range in which `Wind-AE` is physically robust.

(3) The motivation for third point–minimizing hands-on time–is particular to the relaxation method. The relaxation method is incredibly sensitive to being given a good initial ''guess" that is close in parameter space to the goal solution. If the guess is too far from the goal, the relaxation method will not converge. Hence, we provide rampers in the python wrapper that take adaptive and physically-motivated steps through parameters space to allow users to smoothly ramp between solutions across parameters space. For example, from an initial solution for atmospheric escape from a hot Jupiter around a G-star (a Jupiter-sized planet very close to a Sun-like star), `Wind-AE` can automatically ramp to a super-Earth around an M-dwarf (a planet 2-3 times the size of Earth around a much smaller, dimmer star) allowing for parameter space studies to be easily performed. These rampers are also modular in order to serve the first goal (speed) so that users can ramp only the variables they care about.

We elect for an object-oriented framework for the Python wrapper in order to increase user friendliness by packaging the relaxation functions and the postfacto analysis functions into separate objects so that analysis may be performed on exisitng solution files while the relaxation solver is running a new solution.

Research Impact Statement

To date, `Wind-AE` has been used in 4 publications, 3 in-prep publications, 4 grant proposals, and 2 telescope observing time proposals. This is a high-to-average usage rate for comparable open-source atmospheric escape models and is particularly high considering the duration of time that `Wind-AE` has been public.

`Wind-AE` has been used to: simulate observability of atmospheric escape (as in (Pai Asnodkar et al., 2024)), simulate the evolution of planets' atmospheres over time (as in (Tang et al., 2025)), to investigate open questions about the demographics of exoplanets (as in (Loyd et al., 2025)), and make comprehensive grids of mass loss rates (as in (Broome et al., 2025)), including novel grids with water (Tao et al. (in prep)) and high metallicity atmospheres (Broome et al. (in prep)).

AI Usage Disclosure

No generative AI was used in the writing of this manuscript, the accompanying scientific manuscript, or the current version of `Wind-AE`. Future structural updates to the C source code will likely use generative AI and the generated code will be carefully reviewed line-by-line and the wind solutions generated will be validated against pre-gen-AI solutions. Additionally, the outflow structure of any solution will be carefully analyzed for physicality.

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